The goal of this section is to describe the irreducible representations of G.

To begin, let us find the conjugacy classes in $GL_2(\mathbb{F}_q)$.

Representatives	Number of elements in a conjugacy class	Number of classes
Scalar $\left(\begin{smallmatrix} x & 0 \\ 0 & x \end{smallmatrix}\right)$	1 (this is a central element)	$q-1$ (one for every nonzero x)
Parabolic $\left(\begin{smallmatrix} x & 1 \\ 0 & x \end{smallmatrix}\right)$	q^2-1 (elements that commute with this one are of the form $\left(\begin{smallmatrix} t & u \\ 0 & t \end{smallmatrix}\right)$, $t \neq 0$)	$q-1$ (one for every nonzero x)
Hyperbolic $\left(\begin{smallmatrix} x & 0 \\ 0 & y \end{smallmatrix}\right)$, $y \neq x$	q^2+q (elements that commute with this one are of the form $\left(\begin{smallmatrix} t & 0 \\ 0 & u \end{smallmatrix}\right)$, $t, u \neq 0$)	$\frac{1}{2}(q-1)(q-2)$ ($x, y \neq 0$ and $x \neq y$)
Elliptic $\left(\begin{smallmatrix} x & \varepsilon y \\ y & x \end{smallmatrix}\right)$, $x \in \mathbb{F}_q$, $y \in \mathbb{F}_q^\times$, $\varepsilon \in \mathbb{F}_q \setminus \mathbb{F}_q^2$ (characteristic polynomial over $\mathbb{F}_q$ is irreducible)	q^2-q (the reason will be described below)	$\frac{1}{2}q(q-1)$ (matrices with y and $-y$ are conjugate)

Let us explain the structure of the conjugacy class of elliptic matrices in more detail. These are the matrices whose characteristic polynomial is irreducible over $\mathbb{F}_q$ and which therefore don't have eigenvalues in $\mathbb{F}_q$. Let A be such a matrix, and consider a quadratic extension of $\mathbb{F}_q$, namely, $\mathbb{F}_q(\sqrt{\varepsilon})$, where $\varepsilon \in \mathbb{F}_q \setminus \mathbb{F}_q^2$. Over this field, A will have eigenvalues

$$\alpha = \alpha_1 + \sqrt{\varepsilon}\alpha_2$$

and

$$\overline{\alpha} = \alpha_1 - \sqrt{\varepsilon}\alpha_2,$$